Question:

A randomized double-blind controlled trial is conducted on the efficacy of 2 different ACE-inhibitors. The null hypothesis is that both drugs will be equivalent in their blood-pressure-lowering abilities. The study concluded, however, that Medication 1 was more efficacious in lowering blood pressure than medication 2 as determined by a p-value < 0.01 (with significance defined as $p \leq 0.05$). Which of the following statements is correct?Answer Choices:A. This trial did not reach statistical significance.B. There is a 0.1% chance that medication 2 is superior.C. We can fail to accept the null hypothesis.D. We can accept the null hypothesis.

Reasoning + Answer:

<think>The trial concluded that Medication 1 was more efficacious in lowering blood pressure than Medication 2 with a p-value < 0.01. This means the p-value is less than the significance level (p âī¤ 0.05), indicating statistical significance. The null hypothesis states that both drugs are equivalent. Since the p-value is less than 0.05, we can reject the null hypothesis and conclude that Medication 1 is more efficacious than Medication 2. However, failing to accept the null hypothesis is equivalent to rejecting it, and this is indicated by the p-value < 0.01. The probability of 0.1% (0.001) that Medication 2 is superior is not directly related to the p-value and is not a valid statistical interpretation of the results.</think><answer>C. We can fail to accept the null hypothesis.</answer>